Institute of Science and Technology
B.Sc (Hons.) in CSE 2nd Semester
Model Test 2023
CSE: 510225 (Linear Algebra)
Marks: 80, Time: 3 hours

Answer Any Four questions:

1. a) Define degenerate and non-degenerate linear equation. 2

b) Determine the value of λ such that the following system in unknowns x,y,z has (i) a unique solution, (ii) no solution (iii) more than one solutions:

$$\lambda x + y + z = 1$$
$$x + \lambda y + z = 1$$
$$x + y + \lambda z = 1$$

6

c) Find the non trivial solutions of the following system of linear homogeneous equations: 6

$$x_1 - 2x_2 + 3x_3 + x_4 = 0$$
$$x_1 + x_2 - 3x_3 - x_4 = 0$$
$$2x_1 - x_2 + x_3 - x_4 = 0$$
$$2x_1 + 2x_2 - 5x_3 - 3x_4 = 0$$

d) Prove that $\begin{vmatrix} -a & -b & c & d \\ b & -a & -d & c \\ c & -d & a & -b \\ d & c & b & a \end{vmatrix} = (a^2 + b^2 + c^2 + d^2)^2$ 6

2. a) Define (i) Matrix (ii) Square matrix (iii) Idempotent matrix (iv) Singular matrix. 4

b) If A and B are comparable matrices and A' and B' are the transpose matrices of A and B respectively then prove that (i) $(A')' = A$ (ii) $((A+B)' = A' + B'$ (iii) $(AB)' = B'A'$ 6

c) Solve the following system of linear equation with the help of matrix:- 5

$$x + 2y + 3z + 4 = 0$$
$$2x + 4y + 5z + 7 = 0$$
$$3x + 5y + 6z + 10 = 0$$

d) Show that the matrix-

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$$ is idempotent. 5

3.a) Define linear transformation. Show that sum of two linear transformation is a linear transformation. 2+5=7

b) Let S and T be the linear operators of IR^2 into IR^2 defined by S(x,y)=(3x+2y,-6x+y), T(x,y)=(2x+y,x-y). Find formulae defining proportions S+T, ST,TS, and S^2 . 8

c) Show that the mapping $T: IR^2 \to IR^3$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, x_1)$ is linear. 5

4. a) Define rank and nullity of linear transformation. If $T: U \to V$ is a linear transformation from a finite dimensional vector space U to a vector space V then prove that $din(\operatorname{Im} T) + \dim(\ker T) = \dim U$. 2+6=8

b) Find the rank and the nullity of the linear transformation defined on $IR^3 by$ $T(x, y, z) = (x + 2y - z, 2x + y + z, y - z)$ 6

c) Consider the matrix mapping $A: IR^4 \to IR^3$ where

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 5 & -2 \\ 3 & 8 & 13 & -3 \end{bmatrix}$$

Find a basis and the dimension of (i) the image of A and (ii) the Kernel of A.

5.a) Define matrix representation. $Let \{u_1, u_2, \ldots\ldots u_m\}$ be a basis of u and T be any linear operator of U then for any vector $w \in U$ prove that $[T]_u [W]_u = [T(w)]_u$ 2+6=8

b) Let $T: IR^3 \to IR^3$ be the linear mapping defined by T(x,y,z)=(2x+y,x-y,z). Find the matrix representation of T relative to the usual basis of IR^3. 6

c) The matrix of a linear transformation T on IR^2 relative to the usual basis $\{e_1 = (1,0), e_2 = (0,1)\}$ is

$$\begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

Find the matrix of T relative to the basis $\{(-1,3), (2,0)\}$ 6

6.a) Find the characteristic equation of the matrix 2+6=8

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & 1 \end{bmatrix}$$

and verify Cayley-Hamilton theorem for it.

b) Find the eigen values and the corresponding eigen vectors of the matrix. 12

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{bmatrix}$$

Also verify the Cayley-Hamilton the orem for the matrix A.